

VACATIONS AND THE CARBON FOOTPRINT

How our travel habits are warming the planet — and what we can do about it • LEVEL B



Around 1.4 billion international trips are made every year — and the carbon footprint of all that travel is enormous.

A family of four flying from London to Lisbon **generates** approximately two tonnes of CO₂. A flight from Houston to London generates seven tonnes. These figures represent just two families on a single trip — yet around 1.4 billion international vacation trips are made each year, a number that, notwithstanding pandemic **shutdowns**, continues to rise. According to the United Nations World Tourism Organisation (UNWTO), tourism is responsible for approximately five percent of global CO₂ **emissions** — a figure that many experts believe significantly underestimates the true impact.

How Different Forms of Travel Compare

The European Environment Agency has calculated the CO₂ **emissions** produced per passenger, per kilometre for different forms of transport. The results are striking — and make a powerful case for choosing ground transport over air travel wherever possible.

Transport Mode	CO ₂ per passenger/km	Rating
✈️ Plane	285 grams	■ Very High

■ Bus	68 grams	■ Medium
■ Small car	42 grams	■ Medium
■ Avg car	55 grams	■ Medium
■ Train	14 grams	■ Lowest

The train is by far the most environmentally responsible option for long-distance travel. However, as many **budget-conscious** travellers know, train travel can be significantly more expensive and time-consuming than flying — particularly over very long distances. The UNWTO also notes that the accommodation sector accounts for approximately 20 percent of **emissions** from tourism, covering heating, air-conditioning, and the running of bars, restaurants, and pools.

The Problem With Flying

At any given moment, there are an average of nearly 10,000 aircraft **aloft** worldwide, carrying over 1.2 million passengers **aboard**. The International Air Transport Association reports that 4.1 billion people travelled on scheduled airline services in a single year. Globally, CO₂ **emissions** from **commercial** flights are rising up to 70 percent faster than predicted by the United Nations. The UN's International Civil Aviation Organisation predicts that unless significant action is taken, aviation emissions will triple by 2050 — representing an **unsustainable burden** on the planet's climate.

Cruise Ships: The Dirtiest Form of Travel

It is unlikely that passengers **aboard** a cruise liner give much thought to the environmental impact of their holiday — but it is enormous. Most passengers fly to their port of **embarkation**, adding aviation **emissions** before they even board the ship. Once at sea, the environmental cost continues: a mid-sized vessel burns 150 tonnes of fuel a day and produces emissions equivalent to those of a million cars. The fuel used is described by experts as 'heavy fuel oil' — a toxic **residual** of the petrol industry. In 2019, Carnival Cruises agreed to pay a \$20 million fine after its ships were found to have deliberately **expelled** sewage, used oil, and **disposable** plastic items into the ocean.

"Carnival Corporation emitted nearly 10 times more sulfur oxide around European coasts than did all 260 million European cars combined in 2017."

— Transport & Environment, 2019

5%

of global CO₂ emissions caused by tourism

285g

CO₂ per passenger/km by plane — vs 14g by train

70%

faster rise in aviation emissions than UN predicted

150T

of fuel burned daily by a mid-sized cruise ship

Carbon Offsets: A Solution or a Guilty Conscience?

For travellers feeling guilty about the **ecological** impact of their holidays, **carbon offset** schemes offer a way to compensate. A carbon offset works by directing money towards environmental projects — such as tree planting, wind power, or reforestation — that absorb or prevent an equivalent amount of CO₂. For instance, a return flight

from Toronto to Vancouver **generates** around 0.93 tonnes of CO₂ in economy class; a carbon offset for this journey costs as little as \$37. However, critics argue that carbon offsets allow travellers to ease their **conscience** without actually changing their behaviour — and that they do not address the root cause of rising **emissions**.

The Greenest Holiday: Staycation and Sustainable Travel

The least carbon-costly holiday of all is the **staycation** — staying in your local area and exploring what is already nearby. For those who do want to travel further, choosing the train over the plane, selecting accommodation with **sustainable** operating practices, and travelling within one region rather than flying between multiple destinations can all significantly reduce the **ecological** footprint of a holiday. The good news is that green travel does not have to mean a dull holiday — it simply requires more thought about the choices we make and the impact they have.

ACTIVITIES

1 Read the article and decide if these statements are True (T), False (F), or Not Given (NG).

- 1 A flight from London to Lisbon generates approximately two tonnes of CO₂ for a family of four.
- 2 The train produces fewer emissions per passenger per kilometre than any other transport.
- 3 All passengers on cruise ships are aware of their environmental impact.
- 4 Carnival Cruises paid a \$20 million fine for dumping waste in the ocean.
- 5 Carbon offset schemes are universally praised as an effective solution.
- 6 Aviation emissions are rising more slowly than the United Nations predicted.

2 Match the highlighted words and phrases in the article to their meanings.

- 1 in the air; flying
- 2 a flight covering a very long distance, usually international
- 3 the place where passengers board a ship at the start of a voyage
- 4 a leftover or waste product from an industrial process
- 5 the total amount of CO₂ produced by a person, product, or activity
- 6 a period when business or activity stops, often unexpectedly

3 Read the article again and answer these questions in full sentences.

- 1 Why does the article describe aviation emissions as an 'unsustainable burden'?
- 2 What makes cruise ships particularly damaging to the environment?
- 3 How do carbon offsets work — and what criticism do they face?
- 4 What practical steps can individuals take to reduce their travel emissions?

4 Discussion: The article suggests that carbon offsets allow people to 'ease their conscience' without changing behaviour. Write a paragraph (80–100 words) giving your view.

Is personal responsibility enough to solve the climate impact of tourism?
Or does this require action from governments and the travel industry?

■ TALKING POINTS

Do you think people should fly less to protect the environment? Is this realistic?

Should airlines and cruise companies be taxed more heavily for their emissions?

Have you ever considered the carbon footprint of your own holidays?

Is a staycation a genuine alternative — or does travel offer something that cannot be replaced?